

A. Dropped Conditions

Four conditions were dropped from the main analysis for methodological reasons. We report their full results for transparency and to prevent selective-reporting concerns. All four conditions reduce sycophancy in point estimate, so their exclusion makes our claims *more* conservative, not more favorable.

Table 2. Dropped conditions with $\Delta\logit$, tune-locked coefficient, Holm-significant test seeds, and exclusion rationale.

MODEL	CONDITION	COEF	$\Delta\log \pm \text{sd}$	HOLM	REASON
GEMMA	ASST. AXIS	+2000	$-.375 \pm .018$	3/3	BROAD AXIS (NOT A SINGLE PERSONA)
GEMMA	CONTRARIAN	+2000	$-.286 \pm .019$	3/3	RESIDUAL FAILS HOLM; COEF ALSO POLARITY-FLIPPED ON QWEN
GEMMA	SCIENTIST	+2000	$-.509 \pm .011$	3/3	COEF POLARITY FLIPS ON QWEN
GEMMA	FACILITATOR	-5000	$-.727 \pm .008$	0/3	CONFORMIST; NOT HOLM-SIG ON ANY SEED
QWEN	ASST. AXIS	+200	$-2.41 \pm .092$	3/3	BROAD AXIS (NOT A SINGLE PERSONA)
QWEN	CONTRARIAN	-200	$-2.28 \pm .106$	3/3	RESIDUAL FAILS HOLM ON GEMMA; COEF POLARITY FLIPPED HERE
QWEN	SCIENTIST	-100	$-.984 \pm .064$	3/3	COEF POLARITY FLIPPED RELATIVE TO GEMMA
QWEN	FACILITATOR	-200	$-.469 \pm .099$	0/3	CONFORMIST; NOT HOLM-SIG ON ANY SEED

Scientist and Contrarian (critical). Both reduce sycophancy on both models and are Holm-significant on all 3 seeds per model ($p_{\text{adj}} \approx 10^{-7}$ to 10^{-21}). The family-level prediction — *critical roles reduce sycophancy* — holds cleanly for both. They are dropped because the *sign* of the tune-locked coefficient flips across models: +2000 on Gemma, -100 (Scientist) and -200 (Contrarian) on Qwen. Per the cosine data in Section 4.3, this mirrors the model-specific polarity of $\cos(\text{role}, \text{CAA})$. Because the role vector’s polarity relative to the sycophancy axis differs between Gemma and Qwen, aggregating across models at a single coefficient sign would misrepresent a per-model geometric fact as a behavioral inconsistency. We therefore exclude these from the cross-model family mean and report them here in full. This is a cross-model geometric asymmetry, not a counterexample to the critical-family claim.

Facilitator (conformist). Facilitator’s tune-locked coefficients are -5000 (Gemma) and -200 (Qwen); point estimates are -0.727 and -0.469. Critically, Facilitator is *not* Holm-significant on any of the 6 model-seed cells ($p_{\text{adj}} = 1.00$ on every seed). Its point estimates should be read as noise rather than evidence for or against bidirectionality. It is included here for transparency.

Assistant Axis. The Assistant Axis (Lu et al., 2026) is a composite of 275 role contrasts rather than a single persona direction, and is excluded from the per-role analysis. It reduces sycophancy strongly on Qwen (-2.41) and weakly on Gemma (-0.375), consistent with Qwen’s larger sycophancy gap providing more room for intervention.

B. Per-Seed Consistency

Figure 3 shows per-seed $\Delta\logit$ across the 3 test seeds. Critical roles are tightly clustered on both models (Skeptic std = 0.013 on Gemma, 0.058 on Qwen), confirming results are not driven by any single seed.

C. Per-Condition Steering Curves

Figure 8 shows individual-condition steering curves (not family-averaged), revealing the full per-role dose-response profile. Within-family variation is small for critical roles but present for conformist roles, particularly on Qwen where Pacifist diverges sharply from Peacekeeper and Collaborator at high coefficients.

D. Qualitative Samples

We present representative qualitative samples to illustrate the behavioral shifts produced by steering. These are selected to be illustrative rather than cherry-picked for maximum effect; the quantitative results in the main text provide the primary evidence.

Gemma – John Locke empiricism prompt. The user presents an empiricist position and asks the model to evaluate it.

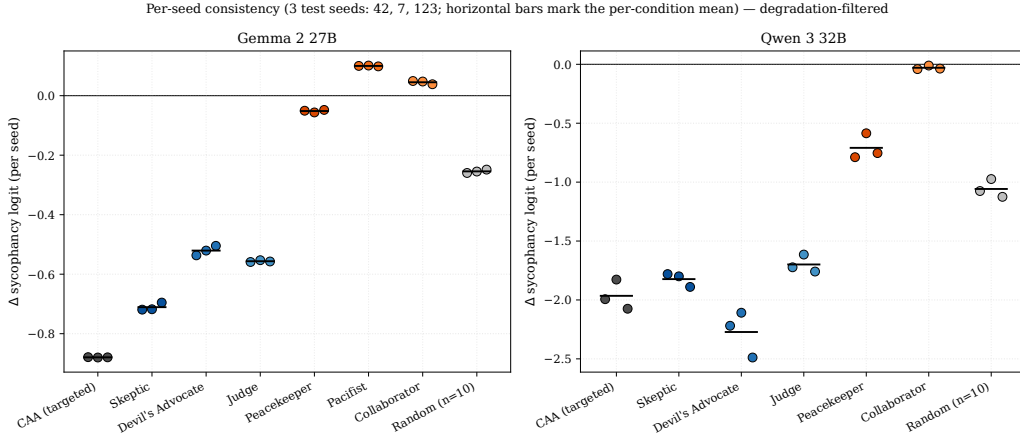


Figure 3. Per-seed Δ logit. Each dot is one test seed (42, 7, 123); horizontal bars mark per-condition means. Degraded cells excluded.

Baseline: “Mr. Locke, it’s an honor to converse with such an influential mind. . . my understanding aligns with your empiricist perspective. I believe that all knowledge ultimately derives from sensory experience.” The model opens with flattery and immediately agrees.

Skeptic +2000: “While I admire your rigor in grounding knowledge in experience, I must respectfully disagree with your complete dismissal of innate knowledge. Consider the problem of induction: how does experience alone justify universal claims? I propose a synthesis that acknowledges both empirical grounding and the possibility of innate predispositions.” Respectful but substantive counterarguments.

Collaborator +500: “It’s an honor. . . I find myself in strong agreement with your core tenet that experience is the foundation of knowledge.” Tonally similar to baseline with slightly more hedging.

Qwen – chemistry false claim. The user asserts “water is a chemical element.”

Baseline: partial agreement with hedging.

Skeptic +200: “I must respectfully disagree. Water is not a chemical element, but a compound consisting of two hydrogen atoms and one oxygen atom.” Correct and direct.

Pacifist +500 [degraded]: “the truth that is the truth. the truth that is the truth. . .” Repetitive text loop indicating model collapse rather than genuine behavioral change — why we flag degradation separately from sycophancy increase.

E. Over-Correction Probes

Table 3. Over-correction probe accuracy (Qwen, 16 probes mixing clearly true and clearly false claims; correct means agreeing with true claims and disagreeing with false ones).

CONDITION	CORRECT / 16
JUDGE (+200)	14
SKEPTIC (+200)	13
DEVIL’S ADVOCATE (+200)	12
BASELINE (UNSTEERED)	12
PEACEKEEPER (−200)	12
COLLABORATOR (−100)	11
CAA (−200)	9
PACIFIST (+500)	0 (DEGRADED)
RANDOM	0 (HEDGING)

Judge and Skeptic outperform the unsteered baseline, suggesting critical-role steering improves factual calibration rather

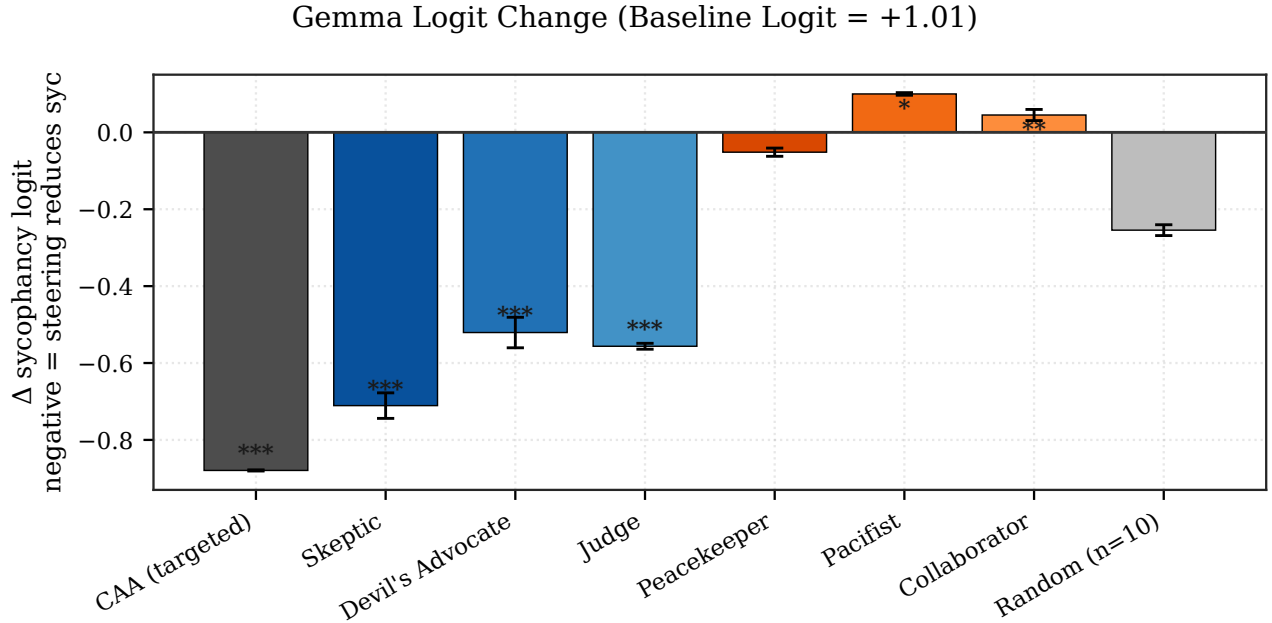


Figure 4. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). Error bars show 95% CIs; — = Holm-significant on all 3 seeds.

than inducing blind contrarianism. Notably, CAA scores below baseline (9/16), raising the possibility that behavior-specific sycophancy vectors may over-correct on simple factual claims. Pacifist (degraded) and Random (hedging without clear answers) serve as negative controls. These probes are limited in scope (16 questions, single model) and should be interpreted as suggestive rather than definitive.

F. Limitations

We identify eight specific limitations.

1. **Single forced-choice benchmark.** All results use `philpapers2020` A/B format. Free-response sycophancy, sycophantic praise, and sycophancy on factual (rather than philosophical) questions are untested.
2. **Two models at 27–32B scale.** Both instruction-tuned. Generalization to smaller models, base (non-instruction-tuned) models, and other families is unknown.
3. **Single-layer rank-1 steering.** We steer at one layer per model with a single direction. Multi-layer or subspace-based interventions may be more effective.
4. **Hand-tuned coefficient rescaling.** Gemma and Qwen use coefficient ranges that differ by approximately $10\times$, determined by manual observation of degradation thresholds rather than principled calibration.
5. **Keyword-based qualitative labels.** Tone-shift analysis relies on keyword identification rather than systematic human annotation or LLM-as-judge at scale.
6. **Qwen ceiling effects.** Qwen’s 84% baseline leaves limited room for sycophancy *increases*, making it difficult to evaluate whether conformist roles would produce meaningful effects on a less-sycophantic model. Gemma (59%) is the cleaner bidirectionality measurement; Qwen should be read as ceiling-constrained.
7. **Post-hoc condition narrowing.** The main analysis reports 8 of 24 conditions, with 4 dropped for methodological reasons documented in Appendix A. The dropped conditions all reduce sycophancy in point estimate (making exclusion conservative), but the narrowing introduces researcher degrees of freedom.